Mtama District Council

PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MTAMA DISTRICT COUNCIL

FORM ONE TERMINAL EXAMINATION
BIOLOGY

Time: 2:30 Hours **MAY, 2026**

STUDENT NAME _____

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of ten (10) questions.
2. Attempt all questions in each section
3. All writing must be in blue or black ink except drawing which must be done in pencil.

QUESTION NUMBER	SCORE	ASSESSOR'S INITIALS
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
TOTAL		
CHECKER'S INITIAL		

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SECTION A: (15 Marks)

Answer all questions in this section.

1. For each item (i–x), choose the most correct answer from the given alternatives and write its letter in the box provided.

   i. During Josiah's study tour to Mikumi National Park, Form One students saw a pisum sativum plant, but he failed to identify the branch of Biology to which it belongs. As a Biology student, identify the correct branch of Biology.

   A. Zoology B. Entomology C. Ecology D. Botany

   ii. _____are conditions or factors that can change or be changed during an experiment.

   A. Hypothesis B. Variables C. Problems D. Conclusion

   iii. The stage in the scientific procedure that is used to compare the results with the hypothesis is known as _____.

   A. Conclusion B. Observation C. Data collection D. Data analysis

   iv. If you work in a field that deals with the health of animals and treatment of their diseases, you are called a _____.

   A. Veterinarian B. Forester C. Agriculturalist D. Pharmacist

   v. A person picks up a telephone after hearing it ring. Which characteristic of living things is shown by this action?

   A. Excretion and reproduction C. Irritability and movement

   B. Respiration and nutrition D. Respiration and growth

   vi. Shany wanted to estimate the population of a certain species of grass in a particular area. Which instrument should be used?

   A. Potter B. Quadrat C. Hand lens D. Microscope

   vii. If the eyepiece lens of a microscope has a magnification of 10× and the objective lens has a magnification of 20×, calculate the total magnification.

   A. 25× B. 200× C. 15× D. 0.2×

   viii. Scientists, engineers, doctors, and geologists use a method to solve scientific problems. This method is called _____.

A. Project work
C. Scientific procedure

B. Data recording
D. Data collection

ix. A statement that summarizes what a scientist has learned from an experiment is called _____.

A. Data interpretation
C. Hypothesis

B. Experimentation
D. Conclusion

x. Which part of the microscope allows the observer to move the body tube up and down in order to obtain a sharp focus of the specimen?

A. Revolving nosepiece
C. Eyepiece lens

B. Fine adjustment knob
D. Coarse adjustment knob

Items number	**i**	**ii**	**iii**	**iv**	**v**	**vi**	**viii**	**ix**	**x**
Answers									

2. Match the functions in list A with the correct part of plant cell responsible in list B by writing it's letter in the space provided.

List A	List B
i. Provide rigidity and maintain the shape of the cell. ii. Control movement of substances in and out of the cell. iii. Site where cellular respiration occurs to release energy. iv. Store cell sap and maintain turgor pressure v. Site where photosynthesis occurs	A. Vacuole B. Cell membrane C. Cell wall D. Cytoplasm E. Nucleus F. Mitochondria G.Chloroplast

List A	i	ii	iii	iv	v
List B					

SECTION B;(70 marks)

Answer all questions from this section

3. In a certain village people face diseases, low crop production and poor waste disposal,One of the biology student related these problem to what they study in class.

   a) From the situation in the village define biology…….

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   b) State four (4) reasons how knowledge of biology is important in solving problems like those observed in the village.
   i)---
   ---
   ii)---
   ---
   iii)---

   ---
   iv)---

   ---

4. A form one student tests whether fertilizer affect plant growth by using two groups of maize plants, one with fertilizer and one without.

   a) i. Define variables_____
   _____
   _____

   b) Mention and explain three (03) types of variables
   i)_____
   _____
   _____
   _____
   ii)_____
   _____
   _____
   iii)
   _____
   _____
   _____

5. A scientist removed mitochondria from the muscles cells of a human and observed that the cells failed to perform their normal activities effectively.

   a) i. Explain the role of mitochondria in the cell._____

   ii. Draw a well labeled structure of mitochondria

   b) Draw a well labeled structure of plant cell

6. Taxonomists discovered several organisms living in a forest,some organisms could mate and produce fertile offspring while others either failed to reproduce or produced infertile offspring.
   a) From above information construct a definition of species_____

b) Design three(3) reasons why organisms from different species do not normally interbreed successfully

i)_____

ii)_____

iii)_____

c) Develop three (3) characteristics used to identify members of the same species .

i)_____

ii)_____

iii)_____

7. In a natural system of classification organisms are grouped based on many features,thus characters that show homology must be distinguished from those that exhibit analogy.

a) With examples describe the meaning of;

i. Homologous characters_____

ii. Analogous characters_____

b) Outline four (4) reasons why natural system of classification is important in classifying organisms?

i)_____

ii)_____

iii)_____

iv)_____

8. A multicellular organisms consists of a few to millions of cells,which have different functions and have features that make them better suited to carry out these functions.

a) Explain the meaning of cell differentiation_____

b) Describe functions of any three specialized plant cells

i)

ii)

iii)

c) i)The following terms are arranged in alphabetical order.Rearrange them into an appropriate biological sequence,starting with the lowest level

Cell → organ → organism → system → tissue

ii)Mention five characterisrics of plant cell

9. A school has decided to convert an empty classroom into a biology laboratory.
   a) What five(5) modification should be made to meet standard laboratory requirements?

b) Write five (5) safety rules that students must obey before using the new laboratory.

SECTION C ;(15 marks)

Answer question number ten (10)

10. Write an essay about photosynthesis using the following guidelines

- Meaning of photosynthesis
- Raw- materials of photosynthesis
- End – product of photosynthesis
- Importance of photosynthesis